



A Three-dimensional Model for the Reversal in the Local Large-scale Interstellar Magnetic Field

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Abstract

We probe the three-dimensional geometry of the large-scale Galactic magnetic field within 1 kpc of the Sun using the Dominion Radio Astrophysical Observatory Global Magneto-Ionic Medium Survey (GMIMS) of the Northern Sky (DRAGONS). DRAGONS is a new full polarization survey of the northern sky from 350 to 1030 MHz covering decl. of $-20^\circ < \delta < 90^\circ$ and a component of GMIMS. The first moment of the Faraday depth (FD) spectra produced from DRAGONS above 500 MHz reveals large-angular-scale FD structures with signs that alternate only once in the southern Galactic hemisphere and twice in the northern hemisphere, patterns shared by other Faraday rotation datasets. DRAGONS is the first survey to achieve high FD resolution while maintaining sensitivity to broad FD structures, enabling the first use of Galactic longitude–FD plots. These plots reveal Faraday-complex structures across the sky, indicating a slablike scenario in which emission and Faraday rotation are mixed. This complexity is overlaid on the same large-scale FD patterns that appear in the first moment map. We model these patterns as a magnetic reversal slicing through the disk on a diagonal and passing above the Sun in Galactic coordinates. We describe this reversal as a plane with a normal vector parallel to the line directed along $(\ell, b) = (168.5, -60^\circ)$ and estimate its distance to be between 0.25 and 0.55 kpc. Our results show that much of the observed Faraday sky may be dominated by the local magnetic field configuration.

Unified Astronomy Thesaurus concepts: Galaxy magnetic fields (604); Milky Way magnetic fields (1057); Interstellar medium (847); Milky Way Galaxy (1054); Radio astronomy (1338); Galaxy structure (622)

1. Introduction

The Galactic magnetic field can be thought of as a large-scale magnetic field, which has coherence lengths on the order of 1 kpc, and small-scale magnetic fields with coherence lengths on the order of a few tens of parsecs (J. C. Brown & A. R. Taylor 2001; T. R. Jaffe et al. 2010; M. Haverkorn 2015; R. Beck 2016). Various models of the large-scale Galactic magnetic field describe the field as generally following the spiral arms, consistent with a logarithmic spiral, where the pitch angle decreases with increasing radius from the Galactic center (e.g., J. C. Brown et al. 2007; X. H. Sun et al. 2008; R. Jansson & G. R. Farrar 2012; J. L. Han 2017).

The “local” large-scale field (within 1 kpc of the Sun) is directed predominantly clockwise as viewed from the north Galactic pole, with a pitch angle of approximately 11.5° (C. L. Van Eck et al. 2011). Models with at least one reversal between the Sun and the Sagittarius Arm, beyond which the

field changes direction and spirals counterclockwise, explain the observed patterns in the rotation measures (RMs) of Galactic pulsars and extragalactic (EG) point sources (e.g., M. Simard-Normandin & P. P. Kronberg 1980; R. C. Thomson & A. H. Nelson 1980; J. L. Han et al. 1997; J. C. Brown et al. 2007; A. P. Curtin et al. 2024). In these observations, the reversal appears in the Galactic plane at longitude $\ell = 60^\circ$, where the line-of-sight (LOS) component of the field appears to abruptly switch direction. The physical location of this reversal is not well known, but is generally assumed to be around 0.5 kpc away in the direction of Galactic longitude $\ell = 0^\circ$ (J. L. Han & G. J. Qiao 1994; C. L. Van Eck et al. 2011), placing it in the local interstellar medium (ISM).

Using data from the Canadian Galactic Plane Survey, A. Ordog et al. (2017; hereafter O17) showed that the position of the reversal in Galactic longitude has latitude dependence and follows a diagonal line. Y. K. Ma et al. (2020) confirmed this observation. O17 proposed that a tilted, plane-like geometry for the reversal could explain their diagonal line, and a recent pulsar study by A. P. Curtin et al. (2024) supported this interpretation. Using pulsar distances to estimate the reversal position in different latitude ranges, they found the reversal is closer at positive Galactic latitudes, and extends farther away into the Sagittarius Arm at negative

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Galactic latitudes. This is what would be expected if the reversal were planar and tilted toward us in the northern Galactic hemisphere.

Much of our understanding of cosmic magnetism is based on observations of Faraday rotation, in which linearly polarized radiation undergoes a rotation of polarization angle when it propagates through a region containing a magnetic field and free electrons. The change in polarization angle, $\Delta\tau$, depends on wavelength squared, λ^2 (units of m^2), as well as the electron number density, n_e (units of cm^{-3}), and the LOS magnetic field component, $B_{\parallel}$ (μG units), along the path, r (in parsecs), as $\Delta\tau = \lambda^2\phi$, where

$$\phi = 0.812 \int_{\text{source}}^{\text{observer}} n_e B_{\parallel} dr \text{ rad m}^{-2}. \quad (1)$$

Depending on the method of measurement, ϕ is referred to either as a Faraday depth (FD) or RM.

Earlier generations of Faraday rotation observations relied on polarization observations at only a small number of radio wavelengths. In these cases, the slope of a linear fit to a plot of τ versus λ^2 gives the RM of the particular LOS. Such a linear fit is only meaningful if there is a single dominant source of polarized emission along the path, Faraday rotating through a foreground screen, a scenario referred to as “Faraday simple.” Bright polarized point sources, such as the active galactic nuclei of external galaxies or pulsars, are excellent examples of this, and numerous RM catalogs have been published using such sources (e.g., R. N. Manchester et al. 2005; J. C. Brown et al. 2007; A. R. Taylor et al. 2009; C. L. Van Eck et al. 2021).

If the emission and Faraday rotation are mixed along the LOS, as in the case of diffuse Galactic synchrotron emission, a τ versus λ^2 plot will not always present as linear. In this case, the LOS ISM is “Faraday complex” as there are multiple ϕ values along the path. An FD spectrum is obtained by applying the Faraday synthesis technique (M. A. Brentjens & A. G. de Bruyn 2005) to observations of polarized emission across a wide, and well-sampled, radio band. For a Faraday-simple LOS, the peak FD and RM are equivalent. For a Faraday-complex LOS, the FD spectrum gives the amount of polarized intensity emitted at each FD.

While the output from Faraday synthesis is a three-dimensional FD cube, correlating FDs with physical distances is not straightforward, especially if there is a magnetic field reversal along the path. Despite this, distance is central to the concept of FD and, therefore, the three-dimensional nature of the FD spectra provides an opportunity to explore the geometry of the large-scale field.

In this work, we adopt the planar geometry suggested by O17 and present a three-dimensional model to describe the magnetic field reversal within the local ISM. In Sections 2 and 3, we describe the data used in our analysis. In Section 4, we identify the observational results that suggest that the FD patterns in our data are connected to the local large-scale magnetic field. Section 5 outlines the mathematical framework for our model of the three-dimensional structure of the reversal in the local ISM. In Section 6, we fit our reversal model to the data, and in Section 7, we discuss the origin of the reversal. Finally, Section 8 offers concluding remarks.

2. Data

We utilize data from the Global Magneto-Ionic Medium Survey (GMIMS) project (M. Wolleben et al. 2010b, 2019, 2021;

Table 1
Parameters of the DRAGONS Data Used in This Paper

DRAGONS Observation Parameters	
Frequency range (MHz)	500–1030
Telescope	DRAO-15
Observation dates	2022 June–2023 January
Angular resolution (deg)	1.3 to 2.45
	All channels convolved to 2.45
Frequency resolution (MHz)	1
Polarization	I , Q , and U
Calibrator	Cygnus A
Sky coverage (deg)	$\delta > -20$
DRAGONS Faraday Synthesis Parameters	
λ^2 range (m^2)	$0.08 < \lambda^2 < 0.36$
$\phi_{\text{resolution}}$ (rad m^{-2})	14
$\phi_{\text{max scale}}$ (rad m^{-2})	37
$\phi_{\text{max detectable}}$ (rad m^{-2})	1327
ϕ_{sampling} (rad m^{-2})	1
CLEAN cutoff σ range (rad m^{-2})	$750 < \phi < 1000$
Sensitivity in FD cube (mK RMSF^{-1})	5

X. Sun et al. 2025), which aims to map diffuse polarized emission across the entire sky over a wide frequency band using large single-dish radio telescopes. When all observations are complete, there will be six GMIMS surveys, with the northern and southern skies each divided into three frequency ranges: low, mid, and high. We use the Dominion Radio Astrophysical Observatory GMIMS of the Northern Sky (DRAGONS) survey (A. Ordog et al. 2026), the low-frequency northern GMIMS survey. We supplement the missing southern decl. in DRAGONS with data from the Southern Twenty centimeter All-sky Polarization Survey (STAPS; N. Raycheva et al. 2025; X. Sun et al. 2025), the high-frequency southern GMIMS survey, which observes 1324–1800 MHz.

2.1. DRAGONS

The DRAGONS survey was observed between 2022 June and 2023 January using the DRAO-15 telescope, an offset paraboloid with an effective diameter of 15 m. Stokes I , Q , and U observations cover frequencies of 350–1030 MHz, in decl. above $\delta = -20^\circ$ over the full range of R.A. The observation strategy and data-processing details are described in A. Ordog et al. (2026). The DRAGONS data used here span the top of the band, from 500 to 1030 MHz. Extending to 350 MHz would have yielded greater resolution in FD, at the expense of a broader beam, and we chose 500 MHz as the lowest frequency as a compromise. Our results are not sensitive to this choice.

To achieve consistent resolution across the DRAGONS survey, the maps were all convolved with a Gaussian kernel to a common angular resolution of $2.45''$, the beam size of the 500 MHz channel. We do not use DRAGONS data within $|b| < 5^\circ$ as they are contaminated by the leakage of high total-power levels into Stokes Q and U near the Galactic plane.

The details of 500–1030 MHz DRAGONS data are summarized in the top half of Table 1. The DRAGONS Stokes I , Q , and U data cubes have 302 frequency channels, each 1 MHz wide, spread across the 500–1030 MHz band. Figures 1(a) and (b) show the DRAGONS frequency coverage and corresponding wavelength squared (λ^2), respectively, with

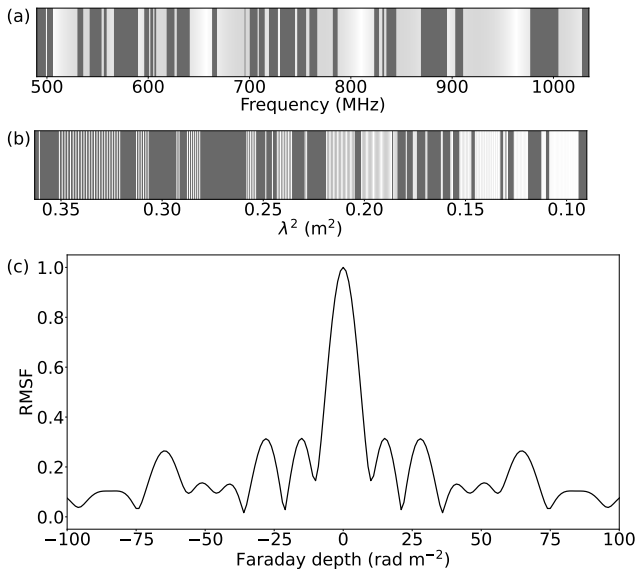


Figure 1. Key DRAGONS parameters. (a) and (b) the DRAGONS frequency and λ^2 coverage, respectively. White lines indicate the 302 frequency channels included in the survey. The dark gaps are due to channels omitted as a result of radio frequency interference. (c) The DRAGONS RMSF.

white bands representing channels included in the data cubes. Frequency channels with significant radio frequency interference contamination in the data were discarded, leaving 57% of the band remaining.

2.1.1. The DRAGONS Faraday Depth Cube

To apply Faraday synthesis to the DRAGONS data, we used the CIRADA RM-Tools package¹² (C. R. Purcell et al. 2020). The 500–1030 MHz DRAGONS bandwidth provides λ^2 coverage from 0.08 to 0.36 m^2 . Due to the finite λ^2 window, the observed FD spectrum is the convolution of the true FD spectrum with the rotation measure spread function (RMSF). The RMSF for DRAGONS is shown in Figure 1(c). The DRAGONS RMSF has an FWHM of $\delta\phi = 14 \text{ rad m}^{-2}$, which determines the FD resolution of the survey. The λ^2 spacing limits the largest FD that can be detected, and the smallest λ^2 determines the maximum width of FD features that can be observed (M. A. Brentjens & A. G. de Bruyn 2005; J. M. Dickey et al. 2019). For DRAGONS, these are $\phi_{\text{max}} = 1327 \text{ rad m}^{-2}$ and $\phi_{\text{max width}} = 37 \text{ rad m}^{-2}$ respectively. As $\phi_{\text{max width}}$ is significantly larger than $\delta\phi$, we can detect wide FD features; this is an exceptional feature of the DRAGONS data, one not achieved in earlier polarization surveys.

The effects of the RMSF side lobe contamination in the FD spectra can be reduced using RM-CLEAN (G. Heald et al. 2009), which is also available in the RM-Tools package. In the first stage of Faraday synthesis, we produced “dirty” (contaminated by the RMSF side lobes) FD spectra out to $\phi = \pm 1000 \text{ rad m}^{-2}$. Based on the FD values observed in other datasets (e.g., GMIMS High-Band North (HBN); M. Wolleben et al. 2021), we did not expect FDs with such high magnitudes; however, the ends of the spectrum were used to assess the RMSF side lobe contribution. Using the method developed by N. Raycheva et al. (2025), we fit a Rayleigh distribution to the polarized intensity for $|\phi| > 750 \text{ rad m}^{-2}$. Then, RM-CLEAN was applied to the dirty spectra and the CLEAN cutoff

threshold for each LOS in the DRAGONS maps set to 3σ , where σ is the width of the fitted Rayleigh distribution.

The DRAGONS FD cube that was prepared for this work has $n_{\text{side}} = 128$ in HEALPix format (K. M. Gorski et al. 2005), with an FD sampling of 1 rad m^{-2} . The properties of these data are summarized in the bottom half of Table 1.

2.2. STAPS

We use STAPS to fill missing information below the southern decl. limit of DRAGONS (i.e., $\delta < -20^\circ$). STAPS was observed with the Parkes Murriyang 64 m telescope and covers the frequency range 1324–1800 MHz (X. Sun et al. 2025). STAPS and DRAGONS have no overlapping frequencies. The higher frequencies of STAPS lead to a poorer FD resolution of $\delta\phi = 140 \text{ rad m}^{-2}$, compared to DRAGONS, which has $\delta\phi = 14 \text{ rad m}^{-2}$. GMIMS Low-Band South (LBS; M. Wolleben et al. 2019), which spans 300–400 MHz, was also an option to fill the missing DRAGONS decl. However, despite being closer in frequency to the DRAGONS band used in this work, LBS is actually farther in λ^2 space than STAPS. For this work, we convolved the STAPS frequency cubes to 2.45 to give these data the same spatial resolution as DRAGONS. We then applied Faraday synthesis to the convolved STAPS cubes, following the processing steps described in N. Raycheva et al. (2025).

2.3. Supporting Datasets

To gain additional insight for our work, we utilized several existing datasets. In particular, we used the G. Edenhofer et al. (2024) three-dimensional dust map and the S. Hutschenreuter et al. (2022) RM map.

2.3.1. G. Edenhofer et al. (2024) Dust Map

Extinction measurements toward stars with parallax distances, determined using the European Space Agency’s Gaia satellite (Gaia Collaboration et al. 2016), have been employed to trace the three-dimensional interstellar dust distribution. These dust maps have emerged as a leading method to explore distances to objects in the local ISM. Several studies have found a correlation between Faraday rotation observations from the Low-Frequency Array Two-meter Sky Survey and maps of interstellar dust (S. Zaroubi et al. 2015; C. L. Van Eck et al. 2017; A. Erceg et al. 2024), demonstrating the potential for using the three-dimensional dust maps to determine distances to Faraday rotation structures.

Here we use the dust map of G. Edenhofer et al. (2024; hereafter E24) to connect DRAGONS FD structures to distances. The E24 map is a statistical model of the interstellar dust distribution produced by applying Bayesian inference to the stellar extinction and Gaia distances of 54 million stars. The result is a three-dimensional map of dust extinction with parsec-scale resolution in logarithmically spaced distance bins from 69 pc to 1.25 kpc.

2.3.2. S. Hutschenreuter et al. (2022) Rotation Measure Map

We use the RM map of S. Hutschenreuter et al. (2022; hereafter H22) as a complementary Faraday rotation dataset to test our magnetic field reversal model. The H22 map was produced using the RMs of 55,190 EG sources, compiled from 41 catalogs, and interpolated using Bayesian inference. The

¹² <https://github.com/CIRADA-Tools/RM-Tools>

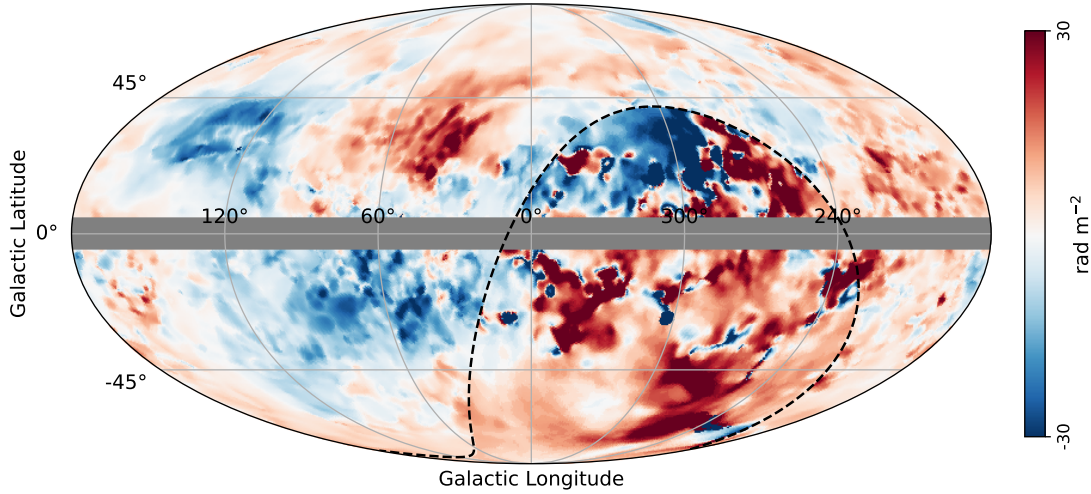


Figure 2. DRAGONS and STAPS FD M1 combined. The STAPS data were convolved to the DRAGONS spatial resolution of 2.45° . The dashed line indicates the boundary between the two datasets. We have masked out the Galactic plane, within $|b| < 5^\circ$, where instrumental effects make the data unreliable.

resulting map describes Faraday rotation across the entire sky along the full path through the Galaxy.

3. The First Moment of the DRAGONS and STAPS Faraday Depth Cubes

Calculation of the first moment (M1) of the FD spectrum enables us to collapse the information contained in three-dimensional FD cubes into two-dimensional maps (J. M. Dickey et al. 2019). For each LOS, M1 is calculated as

$$\text{M1} = \frac{\sum |\tilde{P}_i| \phi_i}{\sum |\tilde{P}_i|}, \quad (2)$$

where $\tilde{P}_i$ is the complex polarized intensity of the i th FD channel, ϕ_i . The combined DRAGONS and convolved STAPS M1 map is shown in Figure 2. FD channels where polarized intensity is below a 6σ threshold were ignored in the summation, where σ is the width of the fitted Rayleigh distribution in each pixel that was determined in the Faraday synthesis stage. For DRAGONS, we calculated the M1 summation in the range $|\phi| < 100 \text{ rad m}^{-2}$ as there is no significant polarized intensity in DRAGONS for FDs beyond this range. For STAPS, we used the full ϕ range, out to $|\phi| < 1000 \text{ rad m}^{-2}$.

The dashed line in Figure 2 indicates the $\delta = -20^\circ$ boundary below which there are no DRAGONS observations and STAPS M1 was used. In the overlap region between the two datasets, $-20^\circ < \delta < 0^\circ$, only DRAGONS values are included, as DRAGONS is the primary dataset being considered in this work. A comparison of STAPS M1 to DRAGONS M1 in the overlap region is shown in Figure 3. In this region, approximately 30% of the STAPS and DRAGONS M1 values are within 10% of each other. For the other 70% of the data points, STAPS M1 tends to be considerably higher in magnitude than DRAGONS M1. This is consistent with visual inspection of Figure 2, where the M1 values are generally darker across the region covered by STAPS. This discrepancy, where STAPS largely presents with higher FDs than DRAGONS, is likely due to depolarization effects.

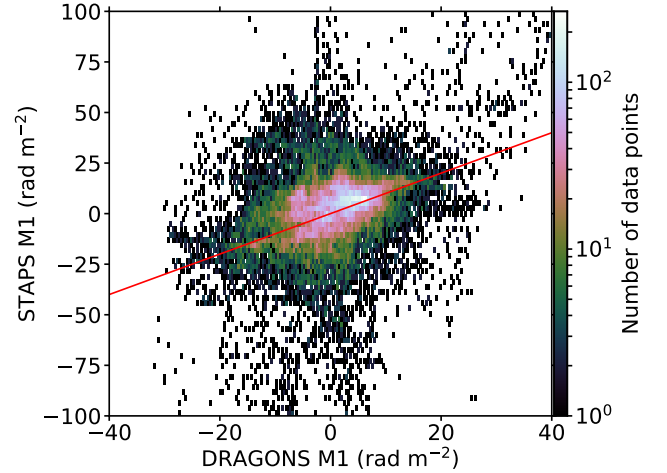


Figure 3. Comparison of STAPS M1 (vertical axis) to DRAGONS M1 (horizontal axis) in the region where the two surveys overlap. The red line is the 1:1 line where STAPS M1 = DRAGONS M1.

3.1. The DRAGONS Polarization Horizon

The resolution of the single dish restricts our observations to diffuse synchrotron emission, and depolarization effects limit the distance probed to a finite polarization horizon (B. M. Gaensler et al. 2001; B. Uyaniker et al. 2003; A. S. Hill 2018). Depolarization arises when a superposition of polarization angles results in lower observed polarized intensity than originally emitted. This may occur if synchrotron emission from different depths is emitted at different angles (geometric depolarization) or experiences varying amounts of Faraday rotation along the path, causing different angles to emerge at the point of observation (depth depolarization). In addition, there may be multiple FDs or emitted polarization angles distributed across the telescope beam (beam depolarization). We expect that depth and beam depolarization are dominant in DRAGONS, as Faraday rotation is enhanced at low frequencies, and the DRAGONS beam is broad.

Through comparisons with pulsar RMs, J. M. Dickey et al. (2019) estimated a polarization horizon averaging around 700 pc for GMIMS HBN (M. Wolleben et al. 2021), which covers

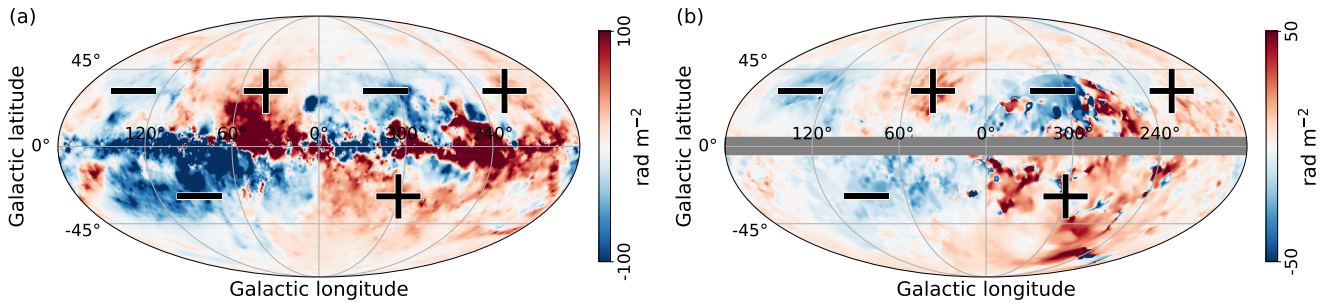


Figure 4. (a) The peaks of the $\sin \ell$ and $\sin 2\ell$ patterns marked with + and – signs plotted on the H22 map. (b) The same + and – peaks marked on the combined DRAGONS and STAPS M1 map.

a frequency range from 1270 to 1750 MHz and has an angular resolution of $40'$. In contrast, for GMIMS LBS, with frequencies spanning 300–480 MHz and an angular resolution of $80'$, they estimated an average polarization horizon around 300 pc. The DRAGONS band lies between the HBN and LBS frequency ranges, and based on depth depolarization, we would expect DRAGONS to be sensitive to emission originating at distances between the HBN and LBS limits, around 500 pc. With its coarser angular resolution, beam depolarization may play a significant role in DRAGONS, and therefore, the average polarization horizon may be even closer. Given this expected nearby polarization horizon, DRAGONS is well suited to explore the local magnetic environment.

Depth depolarization is reduced at the shorter wavelengths of STAPS, and the polarization horizon for that dataset is likely at a greater distance. However, even if STAPS “sees” farther than DRAGONS, the emission recorded in the STAPS data has passed through the same volume that DRAGONS is sensitive to. The Faraday rotation information contained in STAPS is not completely independent of DRAGONS, and therefore, the datasets will still complement each other.

4. Analysis of Observations

In this section, we first identify the large-angular-scale patterns present across the DRAGONS and STAPS M1 maps. Then, we compare the morphology of structures in DRAGONS M1 to regions in the E24 dust map in order to estimate distances to the observed Faraday rotation patterns.

4.1. The Southern $\sin \ell$ and Northern $\sin 2\ell$ Patterns

Many studies of different Faraday rotation datasets have found a common pattern on large angular scales across the sky (e.g., J. L. Han et al. 1997; J. M. Dickey et al. 2022). In midlatitudes in the southern Galactic hemisphere ($b < 0^\circ$), the FDs change signs only once as a function of longitude, appearing as $(- +)$ when centered at $\ell = 0^\circ$, while at midlatitudes in the northern Galactic hemisphere ($b > 0^\circ$), the sign changes twice, appearing as $(- + - +)$. This is shown in Figure 4, which displays these patterns marked on the H22 map and the combined DRAGONS and STAPS M1 maps.

J. M. Dickey et al. (2022) fitted the Fourier sine series

$$\phi = C_0 + C_1 \sin(\ell + \phi_1) + C_2 \sin(2[\ell + \phi_2]), \quad (3)$$

in Galactic longitude, ℓ , to the GMIMS HBN M1 and the H22 map along lines of constant Galactic latitude. They showed that the $\sin \ell$ term dominates the southern $(- +)$ pattern, while

the northern $(- + - +)$ pattern is better described by the $\sin 2\ell$ term.

Examining Figure 4 by eye, the $\sin \ell$ and $\sin 2\ell$ patterns appear to also be present in the DRAGONS and STAPS combined M1 map. While the magnitudes of the FDs are smaller in DRAGONS and STAPS than H22, the positions of the large positive and negative regions appear to peak at roughly the same coordinates.

To mathematically verify this observation, we followed the methods of J. M. Dickey et al. (2022) and used `scipy.optimize.curve_fit` (hereafter `curve_fit`; P. Virtanen et al. 2020) to fit Equation (3) to the DRAGONS and STAPS combined M1 map as a function of Galactic longitude, within 3° wide latitude bins. The fitting results for three northern latitude ranges, with bins centered at $b = 50^\circ, 40^\circ$, and 30° , and three southern latitude ranges, centered at $b = -30^\circ, -40^\circ$, and -50° , are shown in Figure 5. The higher magnitude STAPS M1 values do tend to skew the data in the fourth quadrant, creating uncertainty in the fitting results. However, omitting STAPS and just fitting to DRAGONS led to poor convergence, as `curve_fit` was free to assume any shape across the wide gap where there are no DRAGONS data.

From the `curve_fit` results, we find that the $\sin 2\ell$ coefficient, C_2 , of the fitted sinusoids is approximately twice as large as the $\sin \ell$ coefficient, C_1 , in the northern latitudes. In the southern latitudes, C_1 is consistently more than twice C_2 . This is confirmation that the northern $\sin 2\ell$ and southern $\sin \ell$ patterns are present in both DRAGONS and STAPS.

It is notable that the $\sin \ell$ and $\sin 2\ell$ patterns persist across datasets sensitive to different distance ranges. J. M. Dickey et al. (2022) identified them in GMIMS HBN and H22, and we find them in DRAGONS and STAPS. The most significant difference in how the patterns present in these datasets is the magnitudes of the FDs, suggesting integration along different path lengths through the same large-scale field configuration. As the persistence of this trend across data with different polarization horizons implies a coherence in distance, we consider this evidence that the $\sin \ell$ and $\sin 2\ell$ patterns may result from Faraday rotation through the large-scale Galactic magnetic field. We will explore this possibility in Section 5 with our planar reversal model.

4.2. The Distance to the $\sin 2\ell$ Pattern in DRAGONS

In several magnetic field models, the $\sin \ell$ pattern has been related to the large-scale disk field, and the $\sin 2\ell$ pattern to a more distant field in the Galactic halo, transitioning around a height of $z \approx 0.8$ kpc above the Galactic midplane (e.g., X. H. Sun et al. 2008; M. S. Pshirkov et al. 2011; M. Unger &

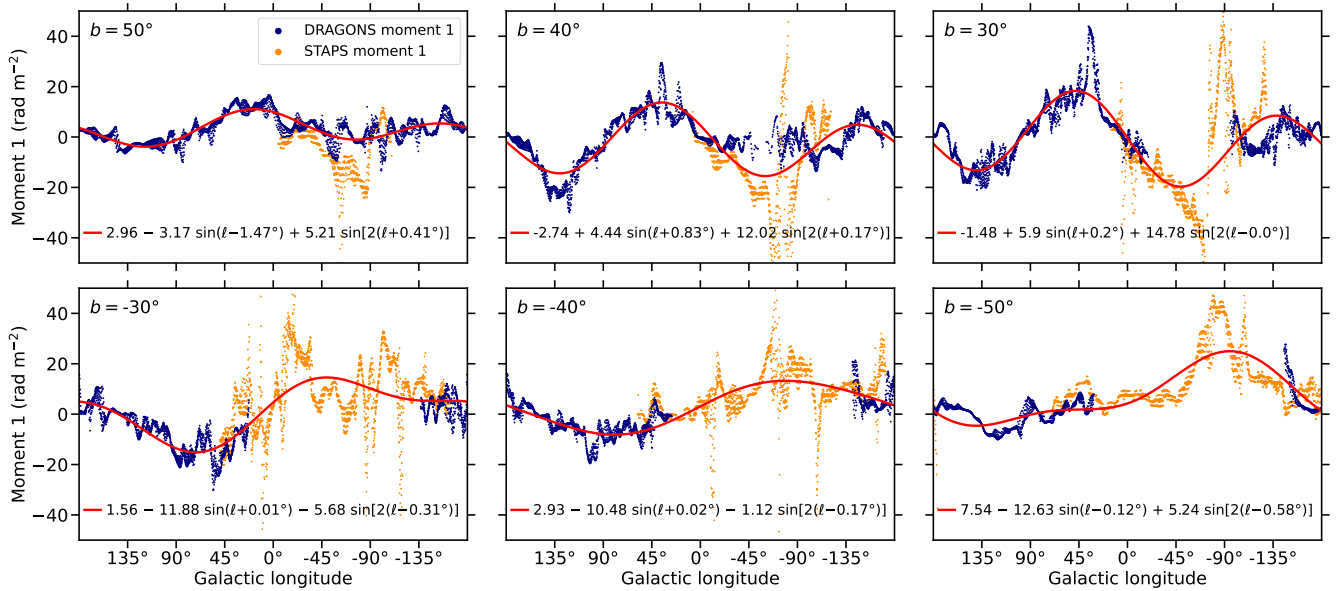


Figure 5. DRAGONS (blue) and STAPS (orange) M1 values plotted along lines of constant latitude. Sinusoids fitted to the combined DRAGONS and STAPS M1 map (Figure 2) along the corresponding latitudes are plotted in red.

G. R. Farrar 2024; A. Korochkin et al. 2025). Others have argued that it is possible for the magnetic field in the shell of the Local Bubble alone to produce such sinusoidal patterns (e.g., E. Maconi et al. 2025). Dust polarization has been shown to correlate with Faraday rotation structures (S. Zaroubi et al. 2015), and G. Halal et al. (2024) found that the magnetic field in dust beyond the Local Bubble contributes to the measured dust polarization fraction on large scales. This suggests that we can expect fields beyond the Local Bubble to contribute to the large-scale FD patterns we observe across the sky. Here, we present evidence that at least some of the magnetic field configuration that produces the $\sin 2\ell$ pattern in DRAGONS is located beyond the Local Bubble, but still within the local ISM.

There are two regions in DRAGONS M1 that stand out as having the largest negative (centered at $\ell = 130^\circ$ and $b = 35^\circ$) and positive (centered at $\ell = 40^\circ$ and $b = 30^\circ$) FDs in the survey. They are positioned at adjacent peaks in the $\sin 2\ell$ pattern, and we show them in a cutout of the DRAGONS M1 map in Figure 6. As Faraday rotation occurs along the entire LOS, out to the polarization horizon, it is not always possible to associate individual distances with FD observations. However, the negative and positive regions identified here have distinct morphologies with clear boundaries. This suggests that magnetic field or electron density enhancements from discrete structures in the ISM may dominate the Faraday rotation along these lines of sight. By identifying the associated ISM structures, we can estimate a distance for these regions.

To identify ISM structures that may be related to the negative and positive M1 regions, we utilized the E24 dust map and applied the histogram of oriented gradients (HOG) method, using the tools available in the AstroHOG library¹³ (J. D. Soler et al. 2019). AstroHOG compares two images by computing the two-dimensional spatial gradient of each and performing a statistical analysis of the distribution of gradient

angle differences across the images, quantifying their morphological agreement.

We rotated the DRAGONS M1 maps to obtain two-dimensional images with 0.5° wide pixels centered on each region of interest to prevent spurious correlations due to projection effects. For the negative region, we centered the images at $(\ell, b) = (130^\circ, 34^\circ)$, and cropped them to 55° by 30° . For the positive region, we centered at $(\ell, b) = (40^\circ, 30^\circ)$, and cropped to 40° by 40° . We then sampled the E24 map out to 1.25 kpc, using the same logarithmically spaced distance bins as E24, and the same Galactic coordinate grids as the DRAGONS M1 images. We used AstroHOG to compare DRAGONS M1 to the E24 map for each distance, using a 2.5° Gaussian kernel. Figures 7(a) and (b) display the projected Rayleigh statistic for the gradient vector direction, V_d (D. L. Jow et al. 2018; J. D. Soler et al. 2025), as a function of distance for the two regions. The V_d statistic is positive when the vector angles in the two images are preferentially more parallel and negative when they are more perpendicular. We also show the Pearson correlation coefficient as a function of distance, which assesses the linear correlation of pixel brightness.

In Figure 7, for both the negative M1 region (a) and the positive M1 region (b), the V_d and Pearson correlation peak together. For the negative region, they have a common peak at $d = 395$ pc, and for the positive M1 region, they peak at $d = 497$ pc. This demonstrates strong morphological agreement between DRAGONS M1 and the dust at these distances, as not only do the pixel brightnesses correlate (Pearson coefficient), the orientations of the shapes in the images also agree (V_d). We show the E24 dust images¹⁴ for these distances in Figures 7(c) and (d). The dust feature in (c), corresponding to the negative FD region, is the North Celestial Pole Loop (NCPL), a well-studied dust and H I cavity (C. Heiles 1989; H. Meyerderks et al. 1991; A. Marchal & P. G. Martin 2023).

¹³ <https://github.com/solerjuan/astroHOG>

¹⁴ We plot $\log_{10}$ of dust extinction, which is recorded by E24 in the units of X. Zhang et al. (2023), and can be translated to extinction at different wavelengths using their extinction curve.

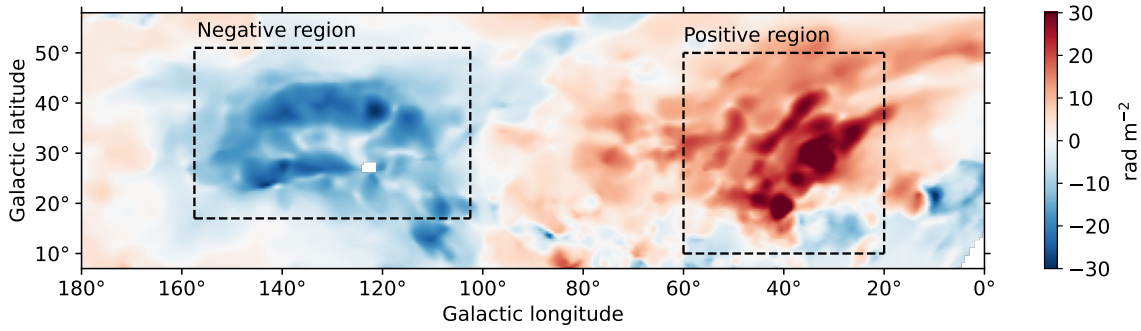


Figure 6. A cutout of DRAGONS M1 displaying the two regions with the highest FDs in the survey.

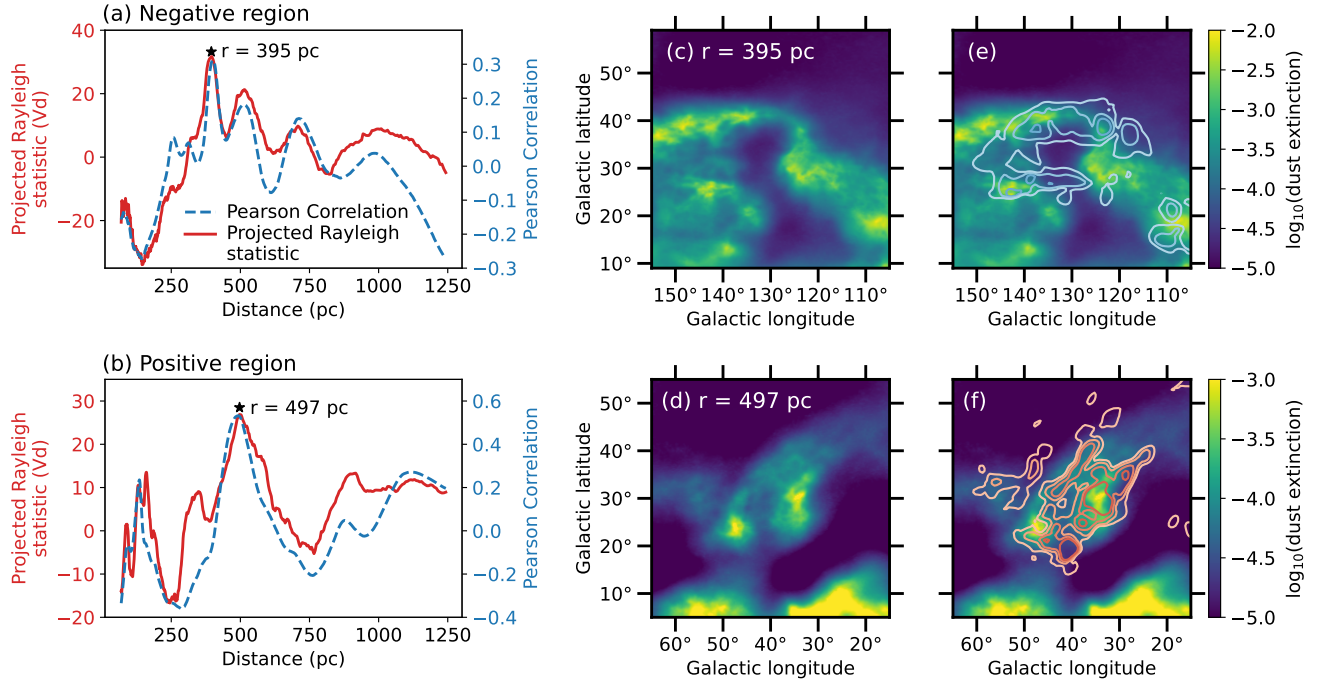


Figure 7. The results of comparing DRAGONS M1 for the two high-Faraday-depth regions to dust at each distance in the E24 map using AstroHOG (J. D. Soler et al. 2019). In (a) and (b) selected statistics as a function of distance are plotted, with the projected Rayleigh statistic (V_d) in red and the Pearson correlation in blue. For both regions, V_d and the Pearson correlation coefficient peak at a preferred distance, marked with a star, and we include the dust map for that distance, with the same map plotted twice in the middle, (c) and (d), and right, (e) and (f), columns. The dust images in (e) and (f) have the DRAGONS M1 contours overlaid, with contours at $|M1| = 15, 20, 25,$ and 30 rad m^{-2} . The panels (c), (d), (e), and (f) approximately cover the regions shown with dashed lines in Figure 6.

The region of strongly positive FDs was identified by M. Wolleben et al. (2010a), who associated it with an H I bubble.

In Figures 7(e) and (f), we include the DRAGONS M1 contours for $|M1| = 15, 20, 25,$ and 30 rad m^{-2} overlaid on the dust images. From this, we are able to see a strong morphological match by eye between the regions in M1 and dust. In both cases, it is notable that some of the M1 structures with the largest FDs match with the dark gaps adjacent to regions of dust (e.g., the arch of positive FD centered at $\ell = 30^\circ$ and $b = 35^\circ$). This is expected, as the dust extinction and Faraday rotation from the same region may not have exact spatial coincidence, but may be instead layered, as different regions of the cloud are exposed differently to sources of ionization. It has been shown that Faraday rotation is particularly sensitive to the ionized and partially ionized ISM phases in ways that other tracers may not be able to detect (e.g., B. Uyaniker et al. 2003; N. Mohammed et al. 2024). The two regions identified here are examples of how the

interconnectedness of the different ISM phases makes multi-ISM tracer analysis important for capturing a complete picture.

In this work, we are interested in determining if a reversal in the local large-scale field can produce the observed $\sin 2\ell$ pattern in the northern sky. The two M1 regions that we have identified here contribute to this pattern, and by associating their Faraday rotation with dust structures, we have estimated their distances to be 395 pc and 497 pc for the negative and positive regions, respectively. At a latitude of $b = 30^\circ$, these distances represent heights of approximately 250 pc above the Galactic midplane, confirming that the associated Faraday rotation occurs within the local disk ISM. From this, we conclude that at least some of the DRAGONS portion of the $\sin 2\ell$ pattern emerges from the local disk.

Some uncertainty does remain as to whether the field direction traced by the regions is connected to the large-scale field. The observed FD enhancements have likely emerged due to conditions local to the regions themselves. The NCPL, for example, has been shown to be host to an unusually strong

magnetic field (C. Heiles 1989), likely resulting from the compression of the ambient magnetic field as the object expanded. However, we argue that their FDs are still informed by the large-scale field in which they are embedded. First, while the two regions have higher FD magnitudes than their surroundings, they share the same sign as the surroundings, suggesting that the ambient LOS field direction was preserved during formation. Second, the FD sign is uniform across each region. If the field lines had been significantly redirected or warped by small-scale perturbations, such as an expanding shock or a dense cloud moving along the LOS, we would expect to see the sign change across the regions as the magnetic field lines that were carried away return to their initial orientation farther along in the ISM. Though we cannot completely eliminate the possibility that the $\sin 2\ell$ pattern is generated by randomly oriented fields due to small-scale perturbations, in the modeling that follows, we demonstrate that a reversal in the local large-scale field is able to reproduce this observed Faraday rotation pattern.

5. A Model for the Reversal in the Local Large-scale Field

The key features of the $\sin \ell$ and $\sin 2\ell$ patterns are the changes in FD sign on large angular scales over the sky. An FD sign change results when the LOS component of the magnetic field changes direction relative to the observer. This may transpire in one of two ways:

1. a geometric sign change occurs when the physical direction of the magnetic field remains the same, but our perspective of it changes as we scan around in Galactic longitude or
2. a magnetic shear across which the magnetic field physically switches direction.

The southern $\sin \ell$ pattern can be explained by the first case, in which the large-scale field spirals clockwise about the Galactic center as viewed from above the north Galactic pole. The LOS component of such a field can be modeled as¹⁵

$$B_{\parallel} = |B| \{ -a \cos b \cos \beta \sin(\ell + p - \alpha) + \sin b \sin \beta \}, \quad (4)$$

where $|B|$ is the total field strength, a is $+1$ (-1) in the case of a clockwise (counterclockwise) field, and p is the field pitch angle. We take the Galactocentric azimuthal angle, α , to be 0° in the direction away from the Sun and increasing in the direction of increasing Galactic longitude. To calculate α , we use a Galactocentric distance r_{GC} of 8.15 kpc (M. J. Reid et al. 2019). The parameter β describes the vertical tilt angle between the horizontal Galactic plane and the field vector.

The $B_{\parallel}$ described by Equation (4) follows a simple $\sin \ell$ relationship, shifting horizontally along the longitude axis with p and α , and with a midline that increases or decreases with latitude, b , and the vertical tilt, β . Thus, the $\sin \ell$ pattern emerges from geometry alone, as we observe the same field from different directions.

The geometric case cannot produce two sign changes around 360° of Galactic longitude. Therefore, a field reversal is

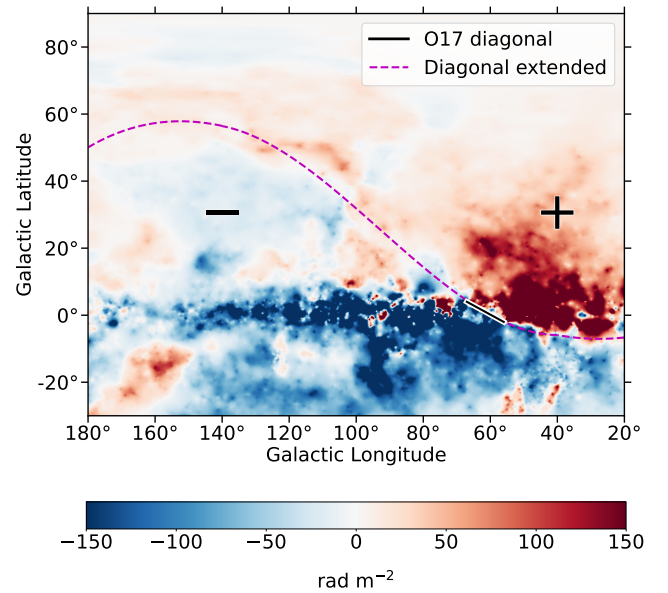


Figure 8. The O17 diagonal (black) plotted on the H22 map. The dashed magenta line demonstrates how a curved extension of the diagonal separates adjacent peaks of the $\sin 2\ell$ pattern, marked with $-$ and $+$ signs.

needed to reproduce the $\sin 2\ell$ pattern in the northern sky. In the case of the large-scale field, such a reversal has been known to exist for decades, having been first identified in pulsar RMs (R. C. Thomson & A. H. Nelson 1980) and EG source RMs (M. Simard-Normandin & P. P. Kronberg 1980). More recently, the three-dimensional and large-scale nature of the reversal was confirmed with a larger set of pulsar RMs by A. P. Curtin et al. (2024). These observations showed that a magnetic shear exists somewhere between the Local Arm and the Sagittarius Arm. We now investigate how this reversal might be extended to higher latitudes within the local disk ISM.

5.1. The Planar Reversal Model

Our model for the reversal is motivated by the work of O17. Using RMs of EG sources as well as Galactic extended emission from the Canadian Galactic Plane Survey, they identified that the reversal follows a diagonal line in Galactic coordinates, extending from $(\ell, b) = (67^\circ, 4^\circ)$ to $(56^\circ, -2^\circ)$. We plot the O17 diagonal as a black line overlaid on the H22 map in Figure 8. Y. K. Ma et al. (2020) extended this investigation into the longitude range 20° – 52° with the addition of 194 new EG source RMs. They suggested that there is an odd-parity disk field in the Sagittarius Arm, switching direction from counterclockwise above to clockwise below the O17 diagonal line.

O17 suggested a planar geometry for the reversal would be consistent with their diagonal. This idea was further supported by A. P. Curtin et al. (2024), who analyzed pulsar RMs and found that the reversal is more nearby in the northern Galactic hemisphere than in the south, where it extends back into the Sagittarius Arm. They describe a potential reversal plane, tilted toward the Galactic center at negative latitudes, to account for this observed configuration. In the local ISM, projection effects are significant, and a plane in physical space would curve when observed in Galactic coordinates. If the O17 line were extended up to higher latitudes and allowed to curve, as illustrated with the dashed magenta line in Figure 8, the

¹⁵ Note that in the coordinate system where the observer is at the origin, as depicted in Figure 9, the LOS component of the magnetic field is positive when directed away from the observer. The convention of integrating from source to observer, as in Equation (1), reverses the FD sign such that FD is positive when the LOS field is directed toward the observer.

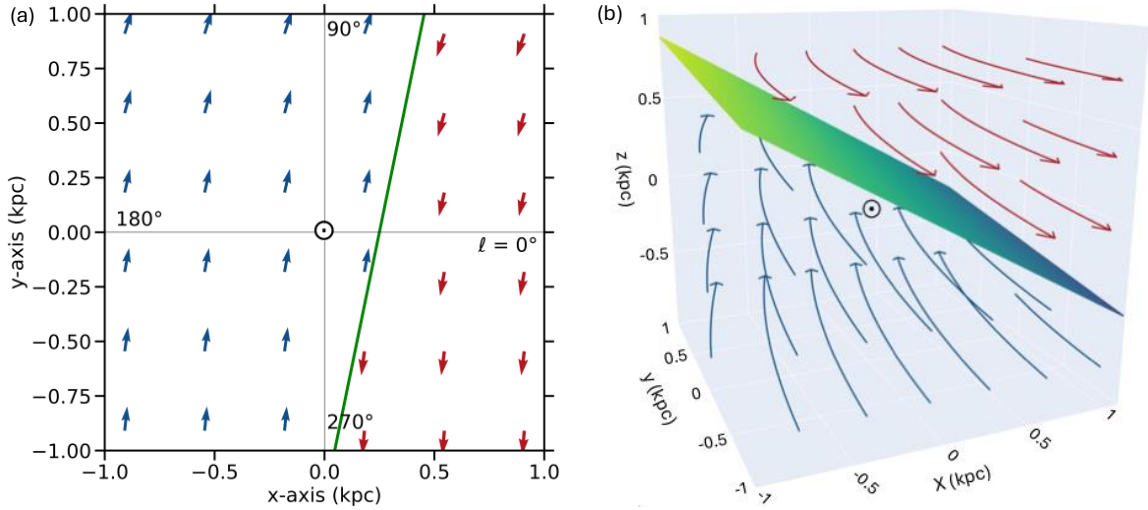


Figure 9. The planar reversal model. (a) A cross section along the x - y plane with the positive x -axis toward the Galactic center, located at (8.15, 0, 0) kpc. The green line indicates the orientation of the reversal, with a clockwise field (blue arrows) on the near side, and counterclockwise field (red arrows) on the far side. (b) The reversal model in three-dimensional space. The reversal is depicted as a diagonal plane separating a region with a clockwise field (blue arrows) below and a counterclockwise field (red arrows) above. The blue to yellow plane colors reflect the z -values to enhance three-dimensional visualization. The position of the Sun is shown by $\odot$.

reversal boundary would separate the adjacent peaks of the $\sin 2\ell$ pattern, marked with $-$ and $+$ signs. This motivates our study of a planar reversal as an explanation for the northern $\sin 2\ell$ pattern.

We model the reversal as a plane,¹⁶ as shown in Figure 9. The reversal plane is angled toward the outer Galaxy in the northern Galactic hemisphere, and toward the Galactic center in the south. Above the plane, the magnetic field is directed counterclockwise, and below, the magnetic field is clockwise. The Sun is positioned below the reversal, in the clockwise region.

We acknowledge that the true nature of the reversal cannot be an infinite flat plane as described here. For example, it is most likely that the reversal curves parallel to the spiral arms. However, near the Sun, this curvature may be considered relatively “flat,” as the Galactocentric azimuthal angle varies by less than 10% within distances inside 1 kpc. Our objective is to test if a plane can successfully approximate the reversal geometry when confined to the narrow “local” region where DRAGONS is most sensitive.

The equation describing the reversal plane is

$$A(x - x_0) + B y + C z = 0, \quad (5)$$

where x_0 is the point where the plane intersects the positive x -axis, in the direction of the Galactic center. The coefficients A , B , and C give the normal vector of the plane, $\hat{n} = A \hat{x} + B \hat{y} + C \hat{z}$, which is directed parallel to the line along Galactic longitude ℓ_n and latitude b_n such that

$$\begin{aligned} A &= \cos b_n \cos \ell_n, \\ B &= \cos b_n \sin \ell_n, \\ C &= \sin b_n. \end{aligned} \quad (6)$$

The parameters that describe the geometry of the reversal plane, the x -intercept, x_0 , and the two tilt angles, ℓ_n and b_n , are depicted in Figure 10.

For any LOS, (ℓ, b) , the distance to the reversal plane is

$$R_p = \frac{A x_0}{A \cos b \cos \ell + B \cos b \sin \ell + C \sin b}. \quad (7)$$

If $R_p < 0$, the LOS passes below the reversal plane and we consider a clockwise large-scale field only. In this case, we calculate the simulated FD for an LOS with no reversal, $\phi_{\text{no-rev}}$, as

$$\phi_{\text{no-rev}}(r) = 0.812 \int_r^0 n_e B_{\parallel \text{CW}} dr', \quad (8)$$

where r is the path length, and $B_{\parallel \text{CW}}$ is given by Equation (4) with $a = +1$. We also use Equation (8) when the path ends before the reversal, i.e., when $r < R_p$.

In the case where an LOS does cross the reversal, we have a clockwise field contribution below, with LOS component $B_{\parallel \text{CW}}$, and a counterclockwise field contribution above, with LOS component $B_{\parallel \text{CCW}}$. For the counterclockwise field, $B_{\parallel \text{CCW}}$ is given by Equation (4) with $a = -1$. We denote the vertical tilt angle below the reversal as β_{CW} and above as β_{CCW} , allowing for the possibility that the vertical field component may behave differently above and below the reversal. To compute the simulated FD for an LOS that crosses the reversal, ϕ_{rev} , we integrate

$$\phi_{\text{rev}}(r) = 0.812 \left\{ \int_r^{R_p} n_e B_{\parallel \text{CCW}} dr' + \int_{R_p}^0 n_e B_{\parallel \text{CW}} dr' \right\}. \quad (9)$$

If models describing the varying electron density and magnetic field strength along the path are included, simulated FDs may be obtained through Equations (8) and (9) by numerical integration. This option remains available for future work. Here, our objective is to demonstrate that a planar reversal can explain the observed large-scale Faraday rotation patterns in the northern sky. To do so, we fit the model to our data, testing many different combinations of model parameters. As numerical integration can be a computationally intensive process, for the purpose of fitting, we apply two approximations to obtain simplified analytical expressions for

¹⁶ A Python library for our planar reversal model is included at https://github.com/beckybooth/Local_reversal.

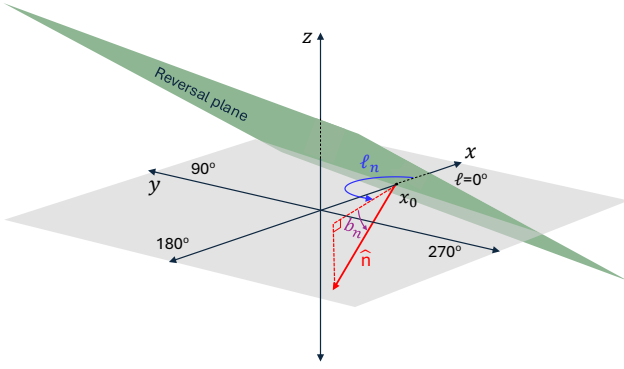


Figure 10. The geometry of the reversal plane model. The Sun is at the origin, and the positive x -axis points in the direction of the Galactic center. The reversal plane intersects the positive x -axis at x_0 . The tilt of the plane is defined by the normal line, $\hat{n}$, which is rotated horizontally by angle ℓ_n and vertically by angle b_n .

the simulated FDs. First, we assume a smooth ISM, with constant electron density and magnetic field strength. These we combine into a single parameter $n_e|B|$, which serves to scale the magnitude of the simulated FDs. We are aware that this approximation does not represent the complexity of the ISM. However, we would require knowledge of the true magnetic field variations and electron density distribution for each LOS to accurately reproduce the complexity we see in the DRAGONS FD spectra. Without this knowledge, a smooth ISM is the simplest approach to demonstrating that our model is able to predict the observed Faraday rotation patterns.

In addition to assuming uniform $n_e|B|$, we approximate the Galactocentric azimuthal angle as $\alpha \approx 180^\circ$. Using this, we integrate Equation (8) and obtain an approximate expression for the FD for LOS with no reversal:

$$\phi_{\text{no-rev}}(r) = \epsilon \zeta r. \quad (10)$$

For an LOS that crosses the reversal, we integrate Equation (9) and have

$$\phi_{\text{rev}}(r) = \epsilon [\zeta R_p + \eta(r - R_p)]. \quad (11)$$

In Equations (10) and (11), for any given LOS toward (ℓ, b) , the constants ϵ , ζ , and η are

$$\begin{aligned} \epsilon &= -0.812 n_e |B|, \\ \zeta &= \cos(b) \cos(\beta_{\text{CW}}) \sin(\ell + p) + \sin(b) \sin(\beta_{\text{CW}}), \text{ and} \\ \eta &= -\cos(b) \cos(\beta_{\text{CCW}}) \sin(\ell + p) + \sin(b) \sin(\beta_{\text{CCW}}). \end{aligned} \quad (12)$$

Maintaining the assumption of uniform $n_e|B|$, these expressions remain valid within $r < 1.5$ kpc. In this distance range, the percent difference between the approximate simulated ϕ values and the results of numerically integrating Equations (8) and (9) through the full range of azimuthal angles remains, on average across the sky, within 5%.

5.2. Simulating Emission: Screens and Slabs

Equations (10) and (11) describe FDs independent of emission and can be utilized in different emission scenarios. The two simplest models for polarized emission are a Faraday screen and a slab. For a screen, all emission is behind the Faraday rotating medium. In this case, the FD spectrum is simple, with a single peak, and M1 is equal to the FD of that

peak. In the case of a screen, for an LOS that does not cross the reversal, the simulated M1 is

$$M1_{\text{screen no-rev}} = \phi_{\text{no-rev}}(R) = \epsilon \zeta R, \quad (13)$$

and for an LOS that does cross the reversal, the simulated M1 is

$$M1_{\text{screen rev}} = \phi_{\text{rev}}(R) = \epsilon [\zeta R_p + \eta(R - R_p)]. \quad (14)$$

Here, we use R to denote the total observed path length. Throughout our exploration of the model, the overall path length, R , remains a free parameter that can be adjusted to fit observations.

For an idealized slab, emission and Faraday rotation are uniformly mixed, and the observed FD spectrum is complex. If there is no reversal along the LOS, M1 is half of the FD of the far end of the slab (B. J. Burn 1966). If there is a reversal, M1 could be any fraction of the farthest FD, depending on at what depth along the slab the magnetic field reverses.

An analytical expression for M1 of a slab can be obtained by integrating Equations (10) and (11) through all distances to emission along the LOS. For a slab, we assume uniform polarized intensity at all distances. In this case, the simulated M1 can be determined from

$$M1_{\text{slab}} = \frac{\int_0^R \phi(r) dr}{\int_0^R dr}. \quad (15)$$

If the LOS does not cross the reversal within the path length, R , we integrate Equation (10), and the simulated M1 is

$$M1_{\text{slab no-rev}} = \frac{\int_0^R \phi_{\text{no-rev}}(r) dr}{\int_0^R dr} = \frac{1}{2} \epsilon \zeta R. \quad (16)$$

If the LOS does cross the reversal, in the numerator of Equation (15) we integrate Equation (10) through distances up to the reversal, R_p , and add this to Equation (11) integrated from R_p to the total path length, R . This gives the simulated M1 for a slab that passes through the reversal as

$$\begin{aligned} M1_{\text{slab rev}} &= \frac{\int_0^{R_p} \phi_{\text{no-rev}}(r) dr + \int_{R_p}^R \phi_{\text{rev}}(r) dr}{\int_0^R dr} \\ &= \frac{\epsilon \left[\zeta \left(R_p R - \frac{1}{2} R_p^2 \right) + \eta \left(\frac{1}{2} R^2 + \frac{1}{2} R_p^2 - R_p R \right) \right]}{R}. \end{aligned} \quad (17)$$

We will fit Equations (13) and (14), the simulated M1 for a screen, and Equations (16) and (17), the simulated M1 for a slab, to DRAGONS M1 in Section 6.

5.3. Reversal Model Parameters

Our planar reversal model has eight parameters. We list these in Table 2. Complex models with large numbers of parameters are prone to overfitting. If a model with fewer parameters can agree with the data, the result is more meaningful and the model is more trustworthy. Thus, we sought to reduce the number of free parameters before fitting our reversal model to the DRAGONS data. To do this, we set the values for the magnetic field pitch angle, p , and the parameters that describe the geometry of the reversal plane, ℓ_n and b_n , and restrict the plane x -intercept distance, x_0 .

Table 2
Reversal Model Parameters

Parameter	Symbol	Value
Electron density and magnetic field strength	$n_e B $	free parameter
Total path length (kpc)	R	<1
Magnetic field:		
Pitch angle (deg)	p	11.5
Tilt below reversal	β_{CW}	free parameter
Tilt above reversal	β_{CCW}	free parameter
Reversal plane:		
Horizontal tilt (deg)	ℓ_n	168.5
Vertical tilt (deg)	b_n	-60
x -axis intercept (kpc)	x_0	0.25–0.55

Literature values for the pitch angle of the large-scale field vary between 0° and 55° (M. Haverkorn 2015). As the field is thought to follow the spiral arms (R. Beck 2016), we use a pitch angle of $p = 11.5$, a value that is used in C. L. Van Eck et al. (2011) and is consistent with the pitch of the Local Arm reported by M. J. Reid et al. (2019).

The horizontal tilt angle of the plane, ℓ_n , can be chosen by considering two requirements. First, in order to maintain $\nabla \cdot \mathbf{B} = 0$, the field lines should not intersect the reversal. This requires that the dot product between the plane normal vector and the magnetic field vanishes. Second, we require that the reversal plane should pass between the Local and Sagittarius Arms, without intersecting either arm. The first requirement leaves ℓ_n relatively flexible in the range 160° – 175° , assuming the vertical tilt angle of field lines remains moderate, within $\beta = \pm 30^\circ$, and $p = 11.5$. To meet the second requirement, we assume the reversal plane runs parallel to the Local Arm in the x - y plane, and set $\ell_n = 168.5$. This ensures that the plane normal is perpendicular to the pitch angle of the Local Arm, as we look toward the Local Arm along the x -axis, and meets the first requirement if β remains small near the reversal plane.

For the vertical tilt of the reversal, b_n , we fit the plane to the O17 diagonal. This RM = 0 rad m⁻² line has a slope in Galactic coordinates of $db/d\ell = 0.55$ and passes through $(\ell, b) = (60^\circ, 0^\circ)$. The line is not visible in DRAGONS M1, either due to obscuration by instrumental effects in the Galactic plane, or because the DRAGONS polarization horizon does not reach far enough for M1 to reach zero in this direction. The observations used by O17 include the RMs of EG sources, which describe Faraday rotation through the entire path through the Galaxy, and thus probe an entirely different distance than DRAGONS. However, the reversal is a local feature, positioned between the Sun and the Sagittarius Arm, consistent with pulsar observations (A. P. Curtin et al. 2024). Since we are modeling a planar reversal, the slope of the reversal observed by O17 would be the same as we observe in DRAGONS. By fitting the slope of the plane to the O17 diagonal, the planar approximation likely best describes the true reversal geometry in the direction of the diagonal, i.e., $(\ell, b) = (60^\circ, 0^\circ)$. If the reversal has any curvature, over longer path lengths the slope would start to deviate from what we have modeled here in directions away from the diagonal. For the short path lengths we use in this work, we are assuming minimal curvature.

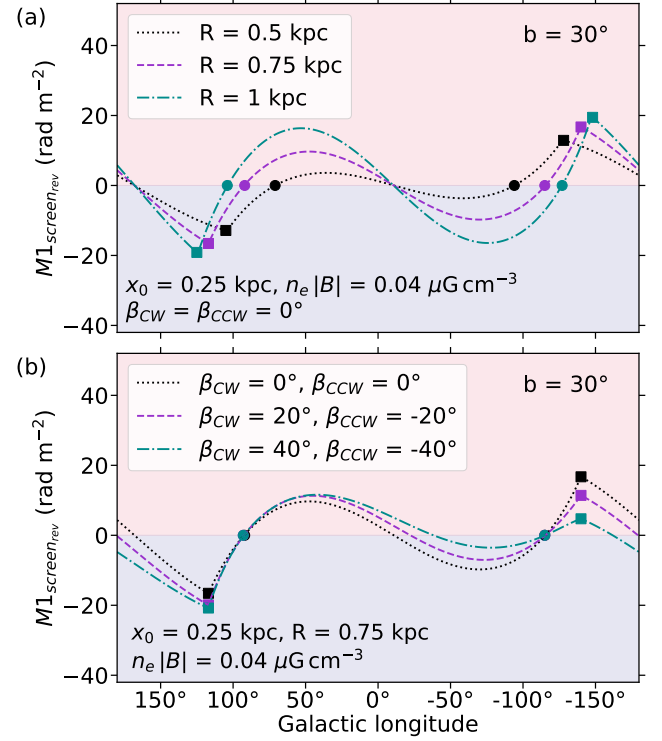


Figure 11. A demonstration of how changing the parameters of the planar reversal model adjusts the simulated $M1_{\text{screen,rev}}$ profiles for Galactic latitude $b = 30^\circ$. Red and blue shading are included for quick visual identification of the positive and negative peaks. (a) Varying the path length, R , shifts the longitude of the zeros (circles) and peaks (squares) horizontally. (b) Varying the vertical tilt angles, β_{CW} and β_{CCW} , shifts the profiles vertically.

Assuming the observations used by O17 have a uniform path length, the field reversal must be at the same distance everywhere along the diagonal line, with the total path length evenly split on either side of the reversal. Taking Galactic latitude, b , to be a function of Galactic longitude, ℓ , and differentiating Equation (7) with respect to ℓ , lines of constant distance to the reversal plane, at $b = 0^\circ$, have slope

$$\left. \frac{db}{d\ell} \right|_{R_p = \text{constant}}^{b=0^\circ} = \frac{\sin(\ell - \ell_n)}{\tan(b_n)}. \quad (18)$$

At $(\ell, b) = (60^\circ, 0^\circ)$, for all reasonable choices of ℓ_n , a slope of $db/d\ell = 0.55$ can be achieved if the vertical tilt of the reversal plane is $b_n \approx -60^\circ$.

We can identify the possible distance range to the reversal based on the dust correlations that we identified in Section 4.2. With $\ell_n = 168.5$ and $b_n = -60^\circ$, the reversal would need to intercept the positive x -axis between $x_0 = 0.25$ and 0.55 kpc in order for it to pass between the NCPL and the M. Wolleben et al. (2010a) bubble.

6. Fitting the Reversal Model to DRAGONS Moment 1

In this section, we will show that our reversal model is able to reproduce the large-scale FD patterns across DRAGONS and identify the best-fit parameter set, informed by the DRAGONS data. First, we demonstrate that our model predicts the northern $\sin 2\ell$ pattern. Figure 11 shows M1 profiles at a latitude of $b = 30^\circ$ for the screen model. We note that varying the free parameters modifies the simulated M1 profiles, but the $\sin 2\ell$ pattern persists. In Figure 11(a), we vary

Table 3
Best-fit Parameters for the Reversal Plane Simulated M1 to DRAGONS M1

Latitude	Emission Model	R/x_0	$n_e B R$ ($\text{cm}^{-3} \mu\text{G kpc}$)	β_{CW} (deg)	β_{CCW} (deg)	Pearson Correlation
All sky	screen	3.0	0.02	18	−23	0.60
	slab	5.1	0.03	18	−23	0.60
−50°	screen	...	0.01	26	...	0.53
	slab	...	0.02	26	...	0.53
−40°	screen	...	0.02	15	...	0.79
	slab	...	0.03	15	...	0.79
−30°	screen	...	0.02	−6	...	0.74
	slab	...	0.03	−6	...	0.74
30°	screen	4.0	0.03	23	−20	0.78
	slab	6.7	0.07	21	−18	0.79
40°	screen	2.1	0.04	17	−12	0.86
	slab	3.5	0.11	15	−10	0.85
50°	screen	1.7	0.03	4	−21	0.69
	slab	2.8	0.09	7	−20	0.71

Note. There is a 10% systematic uncertainty in these fitting results.

the path length, R , and hold the other parameters constant. In Figure 11(b), we vary the tilt angles so that in each case the magnitudes of β_{CW} and β_{CCW} are equal, but the signs are opposite, flipping the tilt angle across the reversal. For the purposes of this demonstration, we use a constant $n_e|B| = 0.04 \mu\text{G cm}^{-3}$ everywhere, based on estimates that $|B| \approx 2 \mu\text{G}$ for the large-scale field (X. H. Sun et al. 2008) and the space-averaged $\langle n_e \rangle \approx 0.02 \text{ cm}^{-3}$ within 1 kpc of the Sun (J. M. Yao et al. 2017).

In Figure 11(a), we observe how increasing path length shifts the position of the zeros and peaks of the simulated M1 profile. There are two zeros that shift with changing R (circles). For the Faraday-screen scenario, these are at the points in the profiles where the total path is roughly evenly split on each side of the reversal, i.e., when $R/R_p \approx 2$. A longer path length moves this point farther along the reversal. The peaks in the profile (squares) are at the points where $R/R_p \approx 1$, and the reversal just enters the LOS. Figure 11(b) shows how changing tilt angles shifts the profiles up or down, but does not significantly change their shapes.

The M1 profiles plotted in Figure 11 are not unique to the parameter sets used. Any parameter set with the same ratio, R/x_0 , and product, $n_e|B|R$, as used in the plots, would produce approximately the same profiles. This is because R and $n_e|B|$ together modulate the amplitude of the peaks, and since x_0 is proportional to R_p (Equation (7)), R/x_0 determines the position of the shifting zeros.

We emphasize that all reasonable variations of the parameters are able to produce the observed northern sky pattern, with two negative M1 peaks and two positive. This demonstrates the robustness of the model, as fine-tuning of the parameters is not required to obtain agreement with the observed large-scale pattern. In Section 4.1, we identified this pattern as having a $\sin 2\ell$ relationship, which is characterized by four evenly spaced peaks. Here, we have shown that the planar reversal model predicts the same two sign reversals with longitude as a $\sin 2\ell$ function, but their spacing is also an important quantity to measure. Next, we seek to fit the parameters such that the amplitude and the spacing of the simulated M1 peaks agree with DRAGONS.

6.1. All-sky Best-fit Parameters

We used `curve_fit` to determine the best-fit parameters that minimize the residuals between the reversal model and DRAGONS M1 data. The goal of this fitting is to demonstrate that the planar reversal geometry that we have modeled here is able to produce M1 maps that are consistent with observations. We fit the reversal model for both kinds of emission scenarios, using the simulated M1 equations for a Faraday screen (Equations (13) and (14)) and a slab (Equations (16) and (17)). Inputs to `curve_fit` were the Galactic coordinates and corresponding M1 values of the DRAGONS M1 HEALPix array. The best-fit model parameters, R/x_0 , $n_e|B|R$, β_{CW} , and β_{CCW} , were determined to minimize the residuals between the M1 data and the models, M1_{screen} and M1_{slab}.

For simplicity, we initially fit one set of parameters for the entire DRAGONS M1 map. We present these fitting results in the top section of Table 3. There is a 10% systematic error in the fitting results, determined by taking twice the square root of the diagonal values of the covariance matrix returned by `curve_fit`. The maps produced from the all-sky fitting are indistinguishable by eye for both emission models. Quantitatively, they differ on average by 4%, which is within the uncertainty introduced by using $\alpha \approx 180^\circ$. We present only the version produced using the best-fit screen parameters in Figure 12, which includes the simulated M1 map for the model (a), and the model plotted on the combined DRAGONS and STAPS M1 map (b). Though STAPS M1 was not used for fitting, we include it to demonstrate that the best-fit model also agrees well with the FD patterns in STAPS.

In the last column of Table 3, we include the Pearson correlation coefficient, which describes the pixel-by-pixel¹⁷ linear correlation, comparing the model to DRAGONS M1. For the all-sky fitting, we find a Pearson correlation of 0.6 in both the screen and slab scenarios. In addition, the best-fit all-sky map has 74% agreement in sign with the DRAGONS M1 by sky area, 70% in the north, and 80% in the south. Both of these indicate good agreement on large scales between the model and the data.

¹⁷ These comparisons are done using $n_{\text{side}} = 128$ HEALPix maps to ensure that the pixels represent roughly equal sky areas.

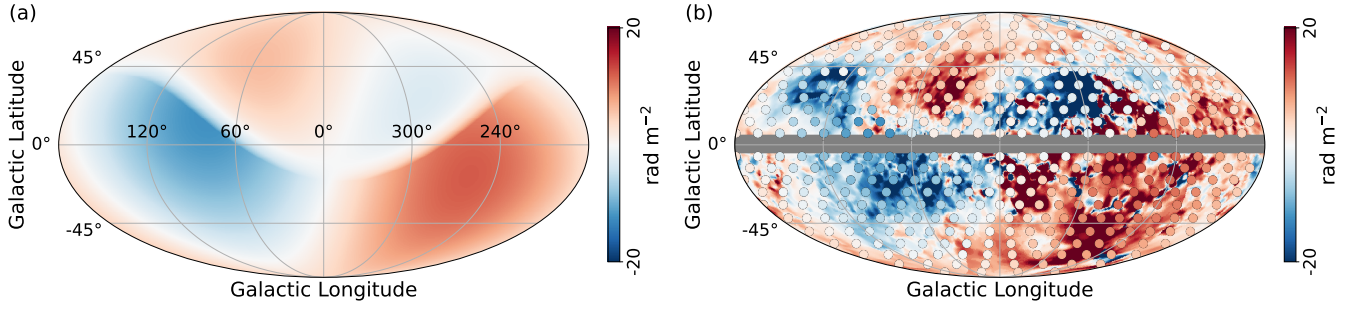


Figure 12. The best-fit simulated M1 map, based on the all-sky best-fit parameters listed in Table 3. (a) The simulated M1 map for the reversal model. As the screen and slab emission models produce nearly identical all-sky maps, this map represents the results produced by both models. (b) The modeled M1 values plotted on the DRAGONS and STAPS combined M1 map.

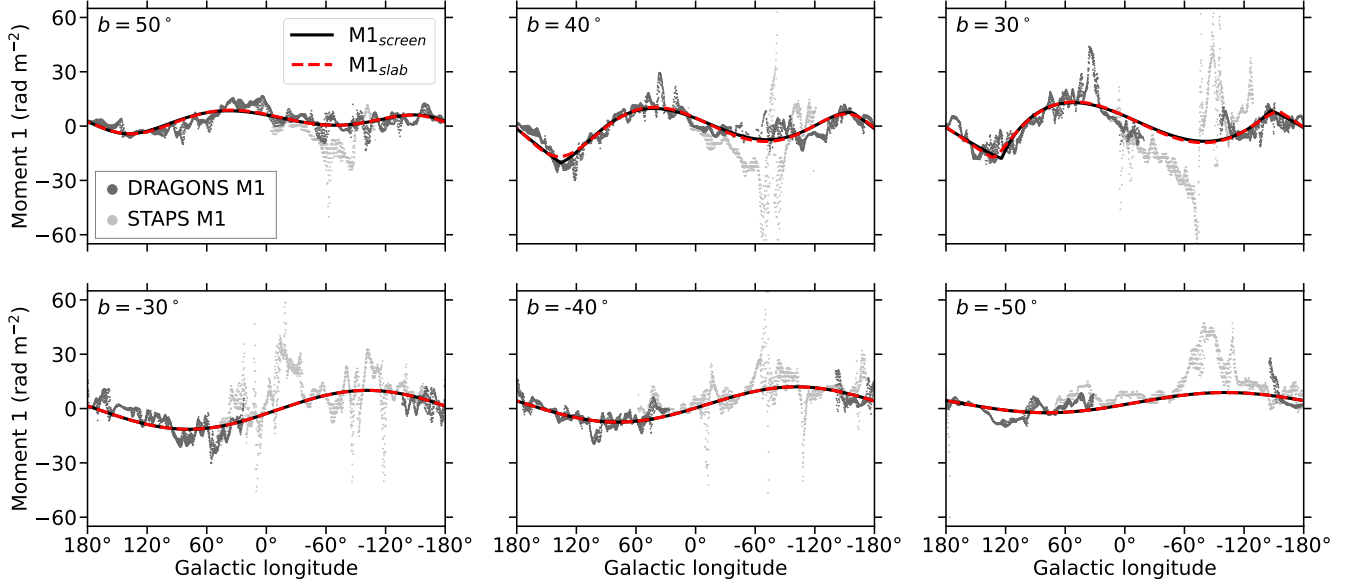


Figure 13. The best-fit modeled M1 profiles fit to DRAGONS M1 within 3° wide bins. The central latitude of the bin is indicated. In each plot, the best-fit screen (black solid line) and slab (red dashed line) models are nearly identical. The models were fit to DRAGONS M1 (dark gray); STAPS M1 values (light gray) are shown for reference only.

Given a reversal distance of $x_0 = 0.25\text{--}0.55$ kpc, the best-fit R/x_0 values predict path lengths ranging from $R = 0.76$ to 1.7 kpc for a screen, and $R = 1.3\text{--}2.8$ kpc for a slab. Aside from the lowest value of $R = 0.76$ kpc, these are beyond what we expect for the DRAGONS polarization horizon. In Section 4.2, we showed that the northern $\sin 2\ell$ pattern is dominated by nearby Faraday rotation, originating between 400 and 500 pc. This discrepancy is likely due to the simplifications we have made to demonstrate the model. While the uniform $n_e|B|$ configuration that we modeled requires farther distances to produce profiles that agree with the data, a nonuniform ISM would be able to generate the same profiles with shorter path lengths.

6.2. Fitting within Latitude Bins

It is unlikely that a single set of parameters can describe the DRAGONS observations across the full sky. In particular, we would expect a varying path length, as the polarization horizon can change with direction. Here, we will explore how the parameters vary with latitude. We fit the reversal model for both screen and slab emission to DRAGONS M1 within 3° wide latitude bins, centered on northern midlatitudes $b = 30^\circ$,

40° , and 50° , and southern midlatitudes $b = -50^\circ$, -40° , and -30° . For each northern latitude, we found best-fit values for the ratio R/x_0 , the product $n_e|B|R$, and the two tilt angles, β_{CW} and β_{CCW} . In the south, only $n_e|B|R$ and β_{CW} were required, as the reversal plane does not extend into the southern midlatitudes that we examined. The fitting results are summarized in the bottom section of Table 3. In all cases, the Pearson correlation is positive, and for most latitudes, the agreement between the model and data is strong, with correlation > 0.7 . This demonstrates that the reversal model is able to produce simulated M1 values that agree very well with observations.

Figure 13 shows the best-fit M1 profiles for both the screen and slab models plotted as a function of longitude on the DRAGONS and STAPS M1 data for each latitude bin. The screen and slab profiles are nearly the same. In both cases, the modeled M1 values are able to reproduce the number of peaks in the observed data, and also the distance between the peaks and the position of the zeros.

Figure 14 shows the best-fit M1 profiles plotted on cross sections of the DRAGONS FD cube, cut along the same latitudes. For these plots, the horizontal axis is Galactic

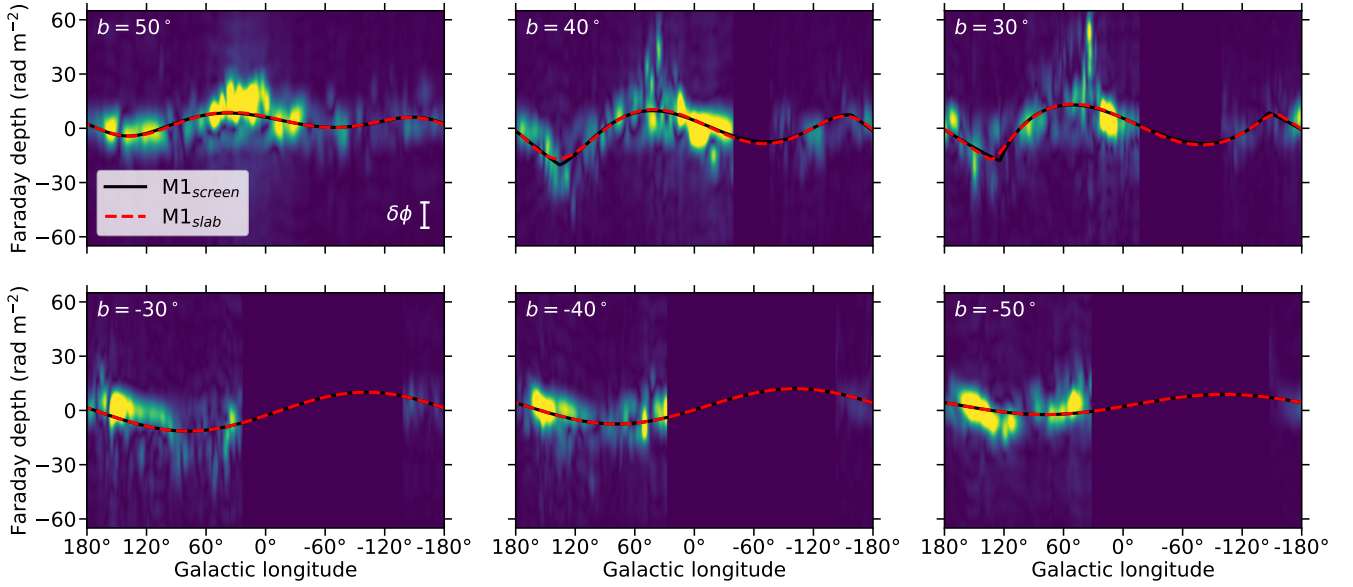


Figure 14. The best-fit models for each latitude range from Figure 13 plotted on the cross section through the DRAGONS FD cube corresponding to that latitude.

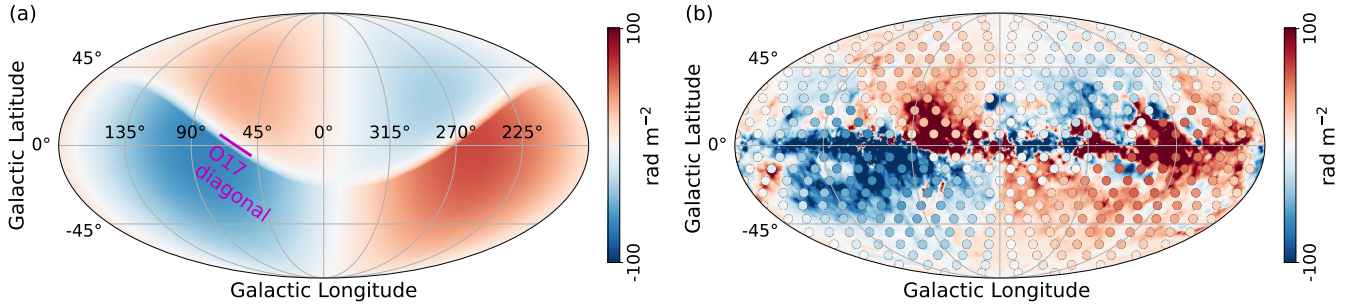


Figure 15. The best-fit simulated RM map for the H22 map. The best-fit parameters are $R/x_0 = 4.5$, $n_e|B|R = 0.07 \text{ cm}^{-3} \mu\text{G kpc}$, $\beta_{\text{CW}} = 0^\circ$, and $\beta_{\text{CCW}} = -7^\circ$. (a) The simulated RM map with the O17 diagonal indicated in magenta. (b) The simulated RM values plotted on top of the H22 map.

longitude and the vertical axis is FD. We refer to them as longitude–FD, or ℓ – ϕ , plots, an analog to the ℓ – ν plots commonly used in spectroscopy. The ℓ – ϕ plots provide a visual representation of how the Faraday complexity in the DRAGONS spectra varies with longitude. They show that the smaller-scale Faraday-complex structures are overlaid on top of the larger-scale coherent pattern predicted by the model and traced by M1 in the data.

The ℓ – ϕ plots also highlight the exceptional nature of the DRAGONS FD data, as we can readily identify emission structures broader than the DRAGONS FD resolution, $\delta\phi$. Broad FD features in the spectra are an indication of “slablike” emission, in which Faraday rotation and emission are mixed, though not necessarily uniformly as in an idealized slab. DRAGONS is the first FD dataset covering large portions of the sky with $\phi_{\text{max width}}$ significantly greater than $\delta\phi$, enabling the detection of slablike emission. J. M. Dickey et al. (2022) showed that the southern sky could be described as slablike, as the southern M1 values observed in GMIMS HBN were approximately one-half of the RMs in the H22 map. They could confirm no such relationship in the north, as is expected when there is a field reversal along the LOS. Using DRAGONS, we can now directly observe the slablike emission in the northern sky through the ℓ – ϕ plots in Figure 14.

6.3. Fitting the Reversal Model to the S. Hutschenreuter et al. (2022) Map

Our simulated M1 map in Figure 12 does not reproduce the O17 diagonal. Instead, the $M1=0 \text{ rad m}^{-2}$ line in Figure 12 passes through the Galactic midplane at $(\ell, b) = (50^\circ, 0^\circ)$. However, as we showed in Figure 11, the zeros of the model change position depending on the path length. As the path length is increased, we expect the 0 rad m^{-2} line to shift higher in longitude. As the O17 diagonal was identified in EG source RMs that were later incorporated into the H22 map, we fit the reversal plane model to the H22 map to show how the model can be adapted to agree with the O17 diagonal.

We used the same fitting procedure as described in Section 6.1. As the H22 map was constructed with EG RMs, only the screen model was required, described by Equations (10) and (11). We fit the model to data in latitudes $|b| > 15^\circ$, to omit the likely more distant and complex structures in the Galactic plane. The resulting best-fit parameters are $R/x_0 = 4.5$, $n_e|B|R = 0.07 \text{ cm}^{-3} \mu\text{G kpc}$, $\beta_{\text{CW}} = 0^\circ$, and $\beta_{\text{CCW}} = -7^\circ$. The simulated RM map for these parameters is shown in Figure 15(a). With the farther path length, the $\text{RM}=0 \text{ rad m}^{-2}$ line now matches the O17 diagonal. Figure 15(b) shows the model values plotted on the H22 map for a visual comparison. The Pearson correlation

comparing the model to the data is 0.46 for the whole sky, and 0.57 for data in latitudes $|b| > 15^\circ$. There is a 74% sign agreement between the map and the model, indicating agreement in the LOS field direction on large angular scales predicted by the model.

6.4. The Vertical Tilt Angle

We find the vertical tilt angle of the field, β , remains robust against the choice of emission model. In our fitting, we found β to be generally positive below the reversal and negative above, with $\beta_{\text{CW}} = 18^\circ$ and $\beta_{\text{CCW}} = -23^\circ$ in the DRAGONS all-sky results. In both cases, this represents a field with a vertical component directed toward the reversal. It is of note that these results are consistent with the magnetic field tilt angles determined using Planck dust polarization observations. In the south, the Planck observations suggest a positive tilt angle of $24^\circ \pm 5^\circ$ (Planck Collaboration et al. 2016), and in the north, they suggest a negative tilt angle of $\sim -18^\circ$ (D. Adak et al. 2020).

7. The Origin of the Reversal

As the reversal is part of the large-scale field, any origin theory should consider the Galactic-scale mechanisms that generate the field as a whole. There are several mechanisms that could lead to the reversal geometry we have identified here. For example, dynamo models, in which kinetic energies of turbulence, convection, and differential rotation shape, amplify, and maintain the Galactic magnetic field, have been shown to predict reversals (e.g., L. Chamandy et al. 2013). J. M. Dickey et al. (2022) presented a number of possible combinations of dipolar and quadrupolar dynamo modes, based on the dynamo models of R. N. Henriksen (2017) and R. N. Henriksen et al. (2018), demonstrating that these combinations can lead to field reversals. It may be possible in future work to identify a combination of dynamo modes that predict our planar reversal geometry.

Another model that gives rise to reversals was described by C. L. Dobbs et al. (2016). They utilized magnetohydrodynamics (MHD) simulations to demonstrate that interaction between the field and spiral arm density shocks can result in reversals. This would explain the spatial correspondence between the reversal and the Local Arm, and indicates that there should be a reversal associated with each arm. Some observations suggest that it is possible for many such reversals to exist (J. L. Han et al. 2018), while others find no evidence of additional reversals outside the solar circle (C. L. Van Eck et al. 2011). Our results agree with C. L. Van Eck et al. (2011) in that even if there are many reversals, only one local reversal is necessary to reproduce a significant portion of the observed Faraday sky.

R. Pakmor et al. (2018) analyzed the magnetic fields of simulated galaxies generated using MHD simulations as part of the Auriga project (R. J. J. Grand et al. 2017). The magnetic fields of the simulated galaxies evolved due to an early turbulent dynamo, followed by a later dynamo phase driven by differential rotation (R. Pakmor et al. 2017). R. Pakmor et al. (2018) produced Faraday rotation maps from different positions within the simulated galaxies. We note in particular that their Figure 8 shows a galactic cross section with a reversal in the azimuthal magnetic field component that appears to follow a planar orientation, much like we have

described here. Moreover, the resulting simulated Faraday rotation maps for a position below that reversal (Figure 9 of R. Pakmor et al. 2018) exhibit the same northern $\sin 2\ell$ -like pattern as observed in our own Galaxy.

By integrating Faraday rotation through different distances in the simulated galaxies, R. Pakmor et al. (2018) demonstrated that the local environment of the observer strongly dominates the resulting map, regardless of the path length chosen. This is particularly true at mid- to high latitudes, where the scale height of the Faraday rotating medium may be relatively close. Our results support this conclusion. In Section 6.3, we were able to reproduce the large-scale Faraday rotation patterns in the H22 map using only our local reversal model and a relatively short path length.

8. Summary

We have utilized the FD data from the DRAGONS survey (A. Ordog et al. 2026), in the frequency range 500–1030 MHz, to test a model for the reversal in the local, large-scale Galactic magnetic field. Here, we define “local” as within a radius of $r = 1$ kpc of the Sun. Our model describes the reversal as a plane positioned between the Sun and the Sagittarius Arm, tilted toward us in the northern Galactic hemisphere, such that the plane passes above us in Galactic coordinates. Below the reversal plane, the large-scale field is directed clockwise, as viewed from the north Galactic pole, and above the reversal, the field is directed counterclockwise. Our key findings are summarized as follows.

1. In Section 4.2, we used the E24 dust map to identify the distance to Faraday rotation structures in the local ISM. By doing this, we have demonstrated the potential for using three-dimensional dust maps to determine Faraday rotation distances. From the identified Faraday rotation structures, we concluded that at least some of the characteristic northern sky ($- + - +$), or $\sin 2\ell$, pattern results from Faraday rotation in the local ISM. In Section 5.3, we used the identified distances to locate the reversal between 0.25 and 0.55 kpc in the direction of the Galactic center.
2. The key feature of our reversal model is the planar geometry. In Section 6, we demonstrated that our planar reversal is capable of reproducing the positive and negative peaks characteristic of the J. M. Dickey et al. (2022) northern $\sin 2\ell$ pattern in both the screen and slab emission scenarios, across various combinations of model parameters. This shows that the $\sin 2\ell$ pattern can be explained simply by a single local field reversal, with no additional magnetic field components required. The planar geometry also explains why no signature of a reversal has been observed in the southern sky, as these lines of sight pass below the reversal.
3. In Figure 14, we presented ℓ – ϕ plots as an effective tool for visualizing FD spectra. The subset of DRAGONS used here has FD resolution of $\delta\phi = 14 \text{ rad m}^{-2}$, while maintaining sensitivity to broader FD features as wide as 37 rad m^{-2} . This was instrumental in making ℓ – ϕ plots possible for the first time. From these plots, we directly observe broadened FD spectra indicating the presence of slablike emission, in which emission and Faraday rotation are mixed, across a significant portion of the sky. The ℓ – ϕ plots also demonstrate how the small-scale

Faraday rotation features, appearing as Faraday complexity in the spectra, are overlaid on top of the large-scale patterns predicted by our reversal model.

4. By fitting our model to the [H22](#) map in Section 6.3, we showed that the planar reversal model can be adapted for data that probe farther distances. As our reversal model only describes a local feature, this supports the idea that Faraday rotation information may be dominated by the local magnetic field configuration, even if the observed emission originates from a greater distance.

While the details of our modeling should be considered in the context of a preliminary model, intended to generate discussion and lay the framework for future investigations, our planar reversal model describes a simple geometry that reproduces much of the large-angular-scale patterns seen in Faraday rotation. We have demonstrated that a full Galaxy model is not necessary to produce the observed Faraday rotation structure; this structure can be reproduced by a single planar reversal integrated through the local volume.

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Facility: DRAO:15m.

Software: AstroHOG (J. D. Soler et al. 2019); Astropy (Astropy Collaboration et al. 2022); healpy (K. M. Gorski et al. 2005; A. Zonca et al. 2019); Matplotlib (J. D. Hunter 2007);

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References

- Adak, D., Ghosh, T., Boulanger, F., et al. 2020, *A&A*, **640**, A100
 Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, *ApJ*, **935**, 167
 Beck, R. 2016, *A&ARv*, **24**, 4
 Brentjens, M. A., & de Bruyn, A. G. 2005, *A&A*, **441**, 1217
 Brown, J. C., & Taylor, A. R. 2001, *ApJL*, **563**, L31
 Brown, J. C., Haverkorn, M., Gaensler, B. M., et al. 2007, *ApJ*, **663**, 258
 Burn, B. J. 1966, *MNRAS*, **133**, 67
 Chamandy, L., Subramanian, K., & Shukurov, A. 2013, *MNRAS*, **428**, 3569
 Curtin, A. P., Weisberg, J. M., & Rankin, J. M. 2024, *ApJ*, **975**, 217
 Dickey, J. M., Landecker, T. L., Thomson, A. J. M., et al. 2019, *ApJ*, **871**, 106
 Dickey, J. M., West, J., Thomson, A. J. M., et al. 2022, *ApJ*, **940**, 75
 Dobbs, C. L., Price, D. J., Pettitt, A. R., Bate, M. R., & Tricco, T. S. 2016, *MNRAS*, **461**, 4482
 Edenhofer, G., Zucker, C., Frank, P., et al. 2024, *A&A*, **685**, A82
 Erceg, A., Jelić, V., Haverkorn, M., et al. 2024, *A&A*, **688**, A200
 Gaensler, B. M., Dickey, J. M., McClure-Griffiths, N. M., et al. 2001, *ApJ*, **549**, 959
 Gaia Collaboration, Brown, A. G. A., Vallenari, A., et al. 2016, *A&A*, **595**, A2
 Gorski, K. M., Hivon, E., Banday, A. J., et al. 2005, *ApJ*, **622**, 759
 Grand, R. J. J., Gómez, F. A., Marinacci, F., et al. 2017, *MNRAS*, **467**, 179
 Halal, G., Clark, S., & Tahani, M. 2024, *ApJ*, **973**, 54
 Han, J. L. 2017, *ARA&A*, **55**, 111
 Han, J. L., & Qiao, G. J. 1994, *A&A*, **288**, 759
 Han, J. L., Manchester, R. N., Berkuijzen, E. M., & Beck, R. 1997, *A&A*, **322**, 98
 Han, J. L., Manchester, R. N., van Straten, W., & Demorest, P. 2018, *ApJS*, **234**, 11
 Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, *Natur*, **585**, 357
 Haverkorn, M. 2015, *ASSL*, **407**, 483
 Heald, G., Braun, R., & Edmonds, R. 2009, *A&A*, **503**, 409
 Heiles, C. 1989, *ApJ*, **336**, 808
 Henriksen, R. N. 2017, *MNRAS*, **469**, 4806
 Henriksen, R. N., Woodfinden, A., & Irwin, J. A. 2018, *MNRAS*, **476**, 635
 Hill, A. S. 2018, *Galax*, **6**, 129
 Hunter, J. D. 2007, *CSE*, **9**, 90
 Hutschenreuter, S., Anderson, C. S., Betti, S., et al. 2022, *A&A*, **657**, A43
 Jaffe, T. R., Leahy, J. P., Banday, A. J., et al. 2010, *MNRAS*, **401**, 1013
 Jansson, R., & Farrar, G. R. 2012, *ApJ*, **757**, 14
 Jow, D. L., Hill, R., Scott, D., et al. 2018, *MNRAS*, **474**, 1018
 Korochkin, A., Semikoz, D., & Tinyakov, P. 2025, *A&A*, **693**, A284
 Ma, Y. K., Mao, S. A., Ordog, A., & Brown, J. C. 2020, *MNRAS*, **497**, 3097
 Maconi, E., Reissl, S., Soler, J. D., et al. 2025, *A&A*, **698**, A84
 Manchester, R. N., Hobbs, G. B., Teoh, A., & Hobbs, M. 2005, *AJ*, **129**, 1993
 Marchal, A., & Martin, P. G. 2023, *ApJ*, **942**, 70
 Meyerdierts, H., Heithausen, A., & Reif, K. 1991, *A&A*, **245**, 247
 Mohammed, N., Ordog, A., Booth, R. A., et al. 2024, *ApJ*, **971**, 100
 Ordog, A., Brown, J. C., Kothés, R., & Landecker, T. L. 2017, *A&A*, **603**, A15
 Ordog, A., Booth, R. A., Landecker, T. L., et al. 2026, *ApJS*, **282**, 53
 Pakmor, R., Gómez, F. A., Grand, R. J. J., et al. 2017, *MNRAS*, **469**, 3185
 Pakmor, R., Guillet, T., Pfrommer, C., et al. 2018, *MNRAS*, **481**, 4410
 Planck Collaboration, Aghanim, N., Alves, M. I. R., et al. 2016, *A&A*, **596**, A105

- Pshirkov, M. S., Tinyakov, P. G., Kronberg, P. P., & Newton-McGee, K. J. 2011, [ApJ](#), **738**, 192
- Purcell, C. R., Van Eck, C. L., West, J., Sun, X. H., & Gaensler, B. M., 2020 RM-Tools: Rotation measure (RM) synthesis and Stokes QU-fitting, Astrophysics Source Code Library,
- Raycheva, N., Haverkorn, M., Ideguchi, S., et al. 2025, [A&A](#), **695**, A101
- Reid, M. J., Menten, K. M., Brunthaler, A., et al. 2019, [ApJ](#), **885**, 131
- Simard-Normandin, M., & Kronberg, P. P. 1980, [ApJ](#), **242**, 74
- Soler, J. D., Beuther, H., Rugel, M., et al. 2019, [A&A](#), **622**, A166
- Soler, J. D., Molinari, S., Glover, S. C. O., et al. 2025, [A&A](#), **695**, A222
- Sun, X., Haverkorn, M., Carretti, E., et al. 2025, [A&A](#), **694**, A169
- Sun, X. H., Reich, W., Waelkens, A., & Enßlin, T. A. 2008, [A&A](#), **477**, 573
- Taylor, A. R., Stil, J. M., & Sunstrum, C. 2009, [ApJ](#), **702**, 1230
- Thomson, R. C., & Nelson, A. H. 1980, [MNRAS](#), **191**, 863
- Unger, M., & Farrar, G. R. 2024, [ApJ](#), **970**, 95
- Uyaniker, B., Landecker, T. L., Gray, A. D., & Kothes, R. 2003, [ApJ](#), **585**, 785
- Van Eck, C. L., Brown, J. C., Stil, J. M., et al. 2011, [ApJ](#), **728**, 97
- Van Eck, C. L., Haverkorn, M., Alves, M. I. R., et al. 2017, [A&A](#), **597**, A98
- Van Eck, C. L., Brown, J. C., Ordog, A., et al. 2021, [ApJS](#), **253**, 48
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, [NatMe](#), **17**, 261
- Wolleben, M., Fletcher, A., Landecker, T. L., et al. 2010a, [ApJL](#), **724**, L48
- Wolleben, M., Landecker, T. L., Hovey, G. J., et al. 2010b, [AJ](#), **139**, 1681
- Wolleben, M., Landecker, T. L., Carretti, E., et al. 2019, [AJ](#), **158**, 44
- Wolleben, M., Landecker, T. L., Douglas, K. A., et al. 2021, [AJ](#), **162**, 35
- Yao, J. M., Manchester, R. N., & Wang, N. 2017, [ApJ](#), **835**, 29
- Zaroubi, S., Jelić, V., de Bruyn, A. G., et al. 2015, [MNRAS](#), **454**, L46
- Zhang, X., Green, G. M., & Rix, H.-W. 2023, [MNRAS](#), **524**, 1855
- Zonca, A., Singer, L., Lenz, D., et al. 2019, [JOSS](#), **4**, 1298